Supplementary Material: RefCompose: Multi-Reference Image Generation via LoRA-Conditioned Diffusion

Anonymous ECCV 2026 Submission

Paper ID #8

A Overview

This supplementary document accompanies the main ECCV workshop paper on **RefCompose**. The main paper focuses on Dense Layout quantitative evaluation on InstanceAssemble [16]; here we provide:

1. **Layout-centric benchmarks** (Section B) comparing the prompt-induced spatial layout stage against dedicated layout generators on COCO-GR [6] numerical and spatial reasoning splits.
2. **Extended ablation study** (Section C) with quantitative metrics complementing the qualitative analysis in the main paper.
3. **Training data details** (Section D), including the synthetic dataset preparation pipeline, referenced from the main experiments section.

Throughout, multiple visual references are composed under a single scene prompt without annotated bounding boxes or per-reference token scaling.

B Layout-Centric Evaluation

We situate the spatial layout stage of RefCompose in the broader layout-generation literature by comparing against LayoutGPT [6] and Lay-Your-Scene [13] on COCO-GR [6] numerical and spatial reasoning splits. RefCompose first induces a coarse composition with a frozen text-to-image model (Flux 2.0 Dev [3]) and then extracts object boxes with Grounding DINO [11].

Metrics. **L-FID** evaluates layout realism by rendering predicted layouts as colour-coded object maps and computing an FID-style distance to ground-truth layouts [7]; lower is better. For *numerical reasoning*, precision, recall, and accuracy measure whether the predicted object set matches the prompt-specified objects. **Numerical GLIP-Accuracy** checks whether the requested object counts appear in the generated image [9]. For *spatial reasoning*, accuracy measures whether predicted boxes satisfy relations such as *left*, *right*, *above*, or *below*; **spatial GLIP-Accuracy** checks whether those relations remain visible after image generation [9].

Table S1: Layout quality on COCO-GR [6] (lower L-FID is better).

Method	L-FID↓
LayoutGPT [6]	3.51
Lay-Your-Scene [13]	3.07
RefCompose (spatial layout)	**2.98**

Table S2: Numerical and spatial reasoning on COCO-GR [6]. GLIP-Accuracy measures image-level count/spatial compliance [9].

	Numerical Reasoning				Spatial Reasoning	
Method	Prec.↑	Recall↑	Acc.↑	GLIP↑	Acc.↑	GLIP↑
LayoutGPT [6]	76.29	86.64	76.72	54.25	87.07	56.89
Lay-Your-Scene [13]	77.62	99.23	95.14	56.20	92.58	58.94
RefCompose (spatial layout)	**78.21**	**99.25**	**96.41**	**56.29**	**93.13**	**59.39**

Results. Table S1 reports layout quality; Table S2 reports numerical and spatial reasoning. RefCompose attains the best L-FID and spatial accuracy among compared methods. We attribute this to inverse layout construction: a large pretrained text-to-image model [3] first produces an aesthetically composed scene, after which boxes are recovered from visible object support. Unlike LLM planners that reason symbolically [6], or layout diffusion transformers trained on narrower layout corpora [13], the T2I prior encodes broad composition statistics from large-scale pretraining; sharper extracted boxes then improve downstream canvas placement and image-level spatial compliance.

C Extended Ablation Study

The main paper presents a qualitative ablation of the reference canvas and Depth Anything 3 [10] depth prior (Figure 7 in the main document). Table S3 reports the corresponding quantitative results on the Dense Layout test split [16], using the same evaluation protocol as Table 2 in the main paper (DINO [5] for composition and reference consistency).

Discussion. Removing the *reference canvas* causes the largest drop on Color, Texture, $\text{DINO}_{\text{refs}}$, and CLIP-I (appearance-centric signals), while Spatial remains relatively high because the depth prior still constrains placement. The model then infers appearance from text alone, landing near the CreatiLayout/InstanceAssemble [14, 16] range on photometric metrics. CLIP-T increases slightly, mirroring the text-alignment trade-off discussed in the main paper.

Removing the *depth prior* causes the largest drop on Spatial and Shape, reflecting degraded placement and occlusion reasoning, while Color, Texture, and $\text{DINO}_{\text{refs}}$ remain closer to the full model since the canvas still supplies appearance. This matches the qualitative observation in the main paper that subjects

Table S3: Ablation study evaluating the contributions of the reference canvas and depth prior. The full model consistently achieves the best performance across identity preservation, spatial arrangement, and semantic alignment metrics.

Configuration	Spatial↑	Color↑	Texture↑	Shape↑	PICK↑	DINO↑	$\text{DINO}_{\text{refs}}$↑	CLIP-I↑	CLIP-T↑
Full model (canvas + depth, dual LoRA)	**0.908**	**0.826**	**0.854**	**0.849**	**0.229**	**0.959**	**0.613**	**0.766**	0.327
w/o reference canvas (text-only appearance)	0.861	0.542	0.571	0.558	0.224	0.712	0.471	0.702	**0.334**
w/o depth prior (canvas only, no structure)	0.683	0.781	0.809	0.612	0.221	0.798	0.587	0.741	0.310

such as the bat and bed remain recognizable, but their arrangement becomes contextually implausible. Together, these results confirm that canvas and depth provide complementary, disentangled signals for appearance and structure.

D Training Data Details

As noted in the main paper, RefCompose is trained on a mixture of a private movie dataset and synthetic composites.

Real data. We curate ultra high-resolution movie frames at 1280×720 from a private dataset sourced from movies. Object references are canonicalized to neutral pose and lighting to reduce spurious correlations between scene context and subject appearance. The five-stage pipeline described in the main paper (Qwen3-VL [1] enumeration, Grounding DINO [11] localisation, Flux 2.0 Dev [3] canonicalization, DINO [5] filtering, and re-annotation) yields approximately 50,000 samples from 1,000 distinct films.

Synthetic data. We complement real cinematic frames with synthetically generated composites that expose the model to diverse spatial arrangements, lighting conditions, object interactions, and viewpoint configurations underrepresented in canonicalized movie crops. The preparation pipeline is described below; together with real data, this mixture encourages generalisation beyond studio lighting and fixed poses without sacrificing photometric fidelity on in-domain cinematic content.

Synthetic dataset preparation pipeline. Our synthetic data pipeline consists of two stages: an upstream *generation* stage and a downstream *conditioning* stage. In the generation stage, we sample three independent external reference images per scene (a background, a face portrait, and an object photo), avoiding self-reconstruction collages by never cropping references from the ground-truth image itself. Each reference is described using Qwen3-VL [1], and the resulting descriptions are composed into a unified scene prompt spanning four scenario categories (normal, lighting, interaction, and position). These prompts, together with the reference images, condition a FLUX.2 model [2] to synthesize the ground-truth target image at 1280×720 resolution. As a final verification step, we use Qwen3-VL [1] as an automatic judge to check whether each sampled reference is faithfully present in the composed target image; samples in which references are missing or inconsistently rendered are discarded, while the remainder are retained for downstream conditioning. Based on empirical insights from this verification stage, we cap the number of composed reference objects per scene at three, which we find sufficient for reliable presence-checking while keeping scene composition tractable. In the downstream stage, we prepare RefCompose training samples by first extracting RGBA cutouts of each verified reference via RMBG-1.4 [4] background removal, then estimating spatial layout through a hybrid approach combining Diffusion Features (DIFT) [15] correspondence matching from a Stable Diffusion [12] U-Net with heuristic bounding-box priors as fallback. These cutouts are composited onto a black canvas at their estimated locations, subject to area constraints (background $\leq$ 100%, person $\leq$ 20%, object $\leq$ 8%), while a monocular depth map is independently predicted from the ground-truth image using Depth Anything 3 [10]. The resulting canvas and depth conditioning signals, alongside the text prompt and ground-truth image, are exported into a training manifest for LoRA [8] fine-tuning.

Training schedule. LoRA [8] training proceeds in three stages over 50,000 iterations at 1280×720, as summarised in the main paper: (1) depth-only warm-up on a black canvas (10,000 iterations); (2) appearance grounding with crops placed at Grounding DINO [11] boxes on ground-truth images (20,000 iterations); (3) robustness fine-tuning with photometric jitter and mild spatial perturbations (remaining iterations). Rank r=32 low-rank adapters are injected into double-stream attention blocks following EasyControl [17].

References

1. Bai, S., Cai, Y., Chen, R., Chen, K., Chen, X., Cheng, Z., Deng, L., Ding, W., Gao, C., Ge, C., Ge, W., Guo, Z., Huang, Q., Huang, J., Huang, F., Hui, B., Jiang, S., Li, Z., Li, M., Li, M., Li, K., Lin, Z., Lin, J., Liu, X., Liu, J., Liu, C., Liu, Y., Liu, D., Liu, S., Lu, D., Luo, R., Lv, C., Men, R., Meng, L., Ren, X., Ren, X., Song, S., Sun, Y., Tang, J., Tu, J., Wan, J., Wang, P., Wang, P., Wang, Q., Wang, Y., Xie, T., Xu, Y., Xu, H., Xu, J., Yang, Z., Yang, M., Yang, J., Yang, A., Yu, B., Zhang, F., Zhang, H., Zhang, X., Zheng, B., Zhong, H., Zhou, J., Zhou, F., Zhou, J., Zhu, Y., Zhu, K.: Qwen3-VL technical report. arXiv preprint arXiv:2511.21631 (2025)
2. Black Forest Labs: FLUX.2: Frontier visual intelligence (2025), `https://bfl.ai/blog/flux-2`
3. Black Forest Labs: FLUX.2 [dev] (2026), `https://huggingface.co/black-forest-labs/FLUX.2-dev`
4. BRIA AI: RMBG-1.4: Background removal model (2024), `https://huggingface.co/briaai/RMBG-1.4`
5. Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P., Joulin, A.: Emerging properties in self-supervised vision transformers. arXiv preprint arXiv:2104.14294 (2021)
6. Feng, W., Zhu, W., Fu, T.j., Jampani, V., Ganin, Y., Yang, X.E., Wang, W.Y.: LayoutGPT: Compositional visual planning and generation with large language models. In: Advances in Neural Information Processing Systems (2024)

7. Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., Hochreiter, S.: GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In: NeurIPS (2017)
8. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: LoRA: Low-rank adaptation of large language models. International Conference on Learning Representations (2022)
9. Li, L.H., Zhang, P., Zhang, H., Yang, J., Li, C., Zhong, Y., Wang, L., Yuan, L., Zhang, L., Hwang, J.N., Chang, K.W., Gao, J.: Grounded language-image pre-training. In: CVPR. pp. 10965–10975 (2022)
10. Lin, H., Chen, S., Liew, J.H., Chen, D.Y., Li, Z., Shi, G., Feng, J., Kang, B.: Depth anything 3: Recovering the visual space from any views. arXiv preprint arXiv:2511.10647 (2025)
11. Liu, S., Zeng, Z., Ren, T., Li, F., Zhang, H., Yang, J., Li, C., Yang, J., Su, H., Zhu, J., et al.: Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. arXiv preprint arXiv:2303.05499 (2023)
12. Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B.: High-resolution image synthesis with latent diffusion models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 10684–10695 (2022)
13. Srivastava, D., Zhang, X., Wen, H., Wen, C., Tu, Z.: Lay-your-scene: Natural scene layout generation with diffusion transformers. arXiv preprint arXiv:2505.04718 (2025)
14. Tang, H., et al.: CreatiLayout: Siamese multimodal diffusion transformer for creative layout-to-image generation. arXiv preprint arXiv:2412.03859 (2024)
15. Tang, L., Jia, M., Wang, Q., Phoo, C.P., Hariharan, B.: Emergent correspondence from image diffusion. In: Advances in Neural Information Processing Systems (2023)
16. Xiang, Q., Sun, S., Li, B., Song, D., Li, H., Chen, N., Tang, X., Hu, Y., Zhang, J.: InstanceAssemble: Layout-aware image generation via instance assembling attention. arXiv preprint arXiv:2509.16691 (2025)
17. Zhang, Y., et al.: EasyControl: Transfer ControlNet to video diffusion for controllable generation and editing. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (2025)